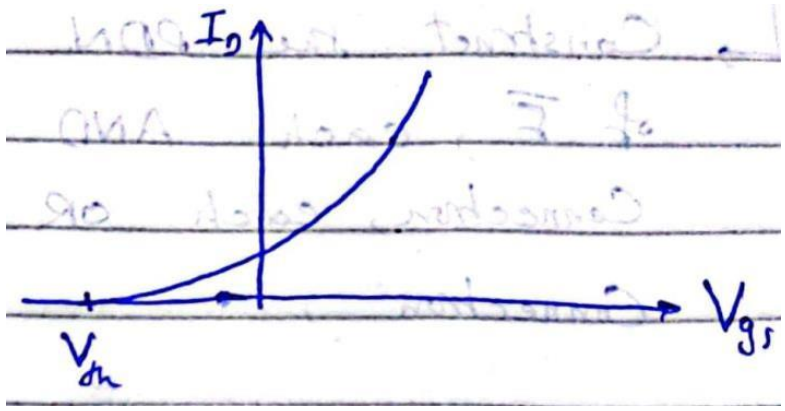
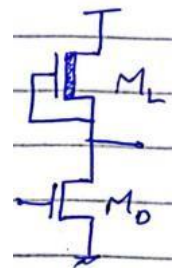
NMOS Depletion-Load Logic Family

- It is a logic family with a depletion NMOS in the load.
- A depletion NMOS is an NMOS with a fabricated channel (i.e. always on). Symbol:

- It has a negative threshold voltage as shown below. This means that there is a channel conducting even for $V_{GS} = 0$.



For the shown inverter:



1) Find V_{OH} and V_{OL}.
2) Draw its VTC showing the noise margins.
3) Find V_{IH} and V_IL.

Answer:

1)

logic input	V_{in}	V_{out}	I	M_D	M_L
'0'	V_{OL}	V_{DD}	0	off	ohmic
'1'	V_{DD}	V_{OL}	I	ohmic	Sat

logic '0' input:

$V_{in} = 0$ & M_D off & $I = 0$. M_L? we are certain that it is on, but sat or ohmic?

By design, we want it in ohmic & assume M_L in ohmic and verify later.

V_{out}? KCL: $I_L|_{ohmic} = I_D|_{off}$

$$k_L\left[(0 - V_{th_L})(V_{DD} - V_{OH}) - \frac{(V_{DD} - V_{OH})^2}{2}\right] = 0 \Rightarrow \boxed{V_{OH} = V_{DD}}$$

M_L is in ohmic

verify: $V_D \overset{?}{<} V_G - V_{th_L} \Rightarrow V_{DD} < V_{DD} - \underset{-ve}{V_{th_L}}$ ✓

logic '1' input:

$V_{in} = V_{DD}$ & M_D on. Sat or ohmic? by design, we want it in ohmic & assume M_D in ohmic

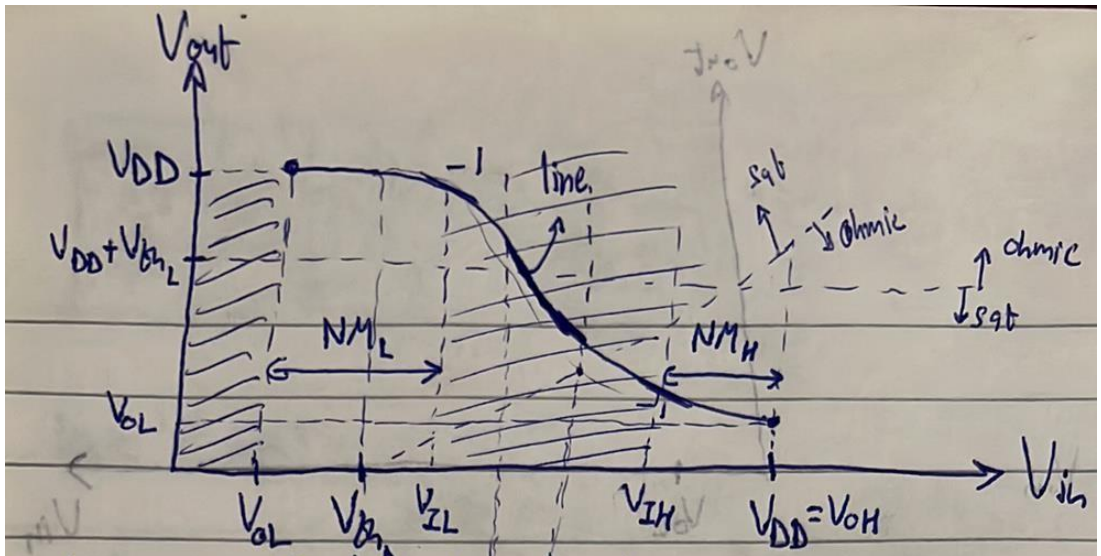
M_L? by design, we want it to be in Sat & assume M_L in sat

V_{out}? KCL: $I_L|_{sat} = I_D|_{ohmic}$

$$\frac{k_L}{2}(0 - V_{th_L})^2 = k_D\left[(V_{DD} - V_{th_D})(V_{OL}) - \frac{V_{OL}^2}{2}\right] \Rightarrow \text{get } V_{OL}$$

verify your assumptions (hena mesh hn3rf ne verify bs fel masa2el hybq m3ak argam fi ht3rf)

2)



M_D: off | sat | ohmic

M_L: ohmic | sat

M_D: $V_{out} > V_{in} - V_{tn_D}$ (sat)

M_L: $V_{DD} > V_{out} - V_{tn_L}$ (sat)

$V_{out} < V_{DD} + \underset{-ve}{V_{tn_L}}$

3) V_{IL}: $I_D|_{sat} = I_L|_{ohmic}$

$$\frac{k_D}{2}(V_{in}-V_{tn_D})^2 = k_L\left[(0-V_{tn_L})(V_{DD}-V_{out}) - \frac{(V_{DD}-V_{out})^2}{2}\right] \rightarrow (1)$$

$$\frac{\partial}{\partial V_{in}}\Big(\quad k_D(V_{in}-V_{tn_D}) = k_L\left[V_{tn_L}\frac{dV_{out}}{dV_{in}} + (V_{DD}-V_{out})\frac{dV_{out}}{dV_{in}}\right]$$

Put $\frac{dV_{out}}{dV_{in}} = -1$: $k_D(V_{in}-V_{tn_D}) = k_L(V_{out}-V_{DD}-V_{tn_L}) \rightarrow (2)$

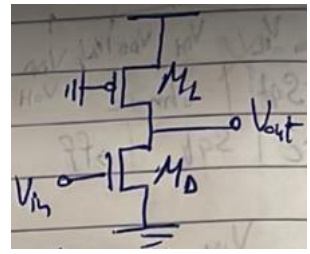
Solve (1) & (2)

Same steps for V_{IH}: I_D|ohmic = I_L|sat

Pseudo-NMOS Logic Family

- It is a logic family with a PMOS in the load having its gate grounded.

For the shown inverter:

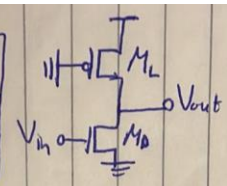


1) Find V_{OH} and V_{OL}.
2) Draw its VTC showing the noise margins.
3) Find V_{IH} and V_{IL}.

Answer:

1)

logic input	V_{in}	M_D	M_L	I	V_{out}
'0'	VOL	off	ohmic	0	V_{DD}
'1'	VDD	ohm	sat	I	VOL



logic '0' input:

$V_{in}=0$ ∴ M_D off & $I=0$. M_L? $V_{SG} \overset{?}{>} |V_{tp}| \Rightarrow V_{DD} \overset{?}{>} |V_{tp}|$ ✓ ∴ on

Sat or ohmic? we want it ohmic & assume ohmic. V_{out}? KCL:

$$I_L|_{ohmic} = I_D|_{off}$$

$$k_L\left[(V_{DD}-0-|V_{tL}|)(V_{DD}-V_{OH})-\frac{(V_{DD}-V_{OH})^2}{2}\right]=0 \Rightarrow V_{OH}=V_{DD}$$

verify that M_L in ohmic: $V_D \overset{?}{>} V_G+|V_{tL}| \Rightarrow V_{DD} > 0+|V_{tL}|$ ✓

logic '1' input:

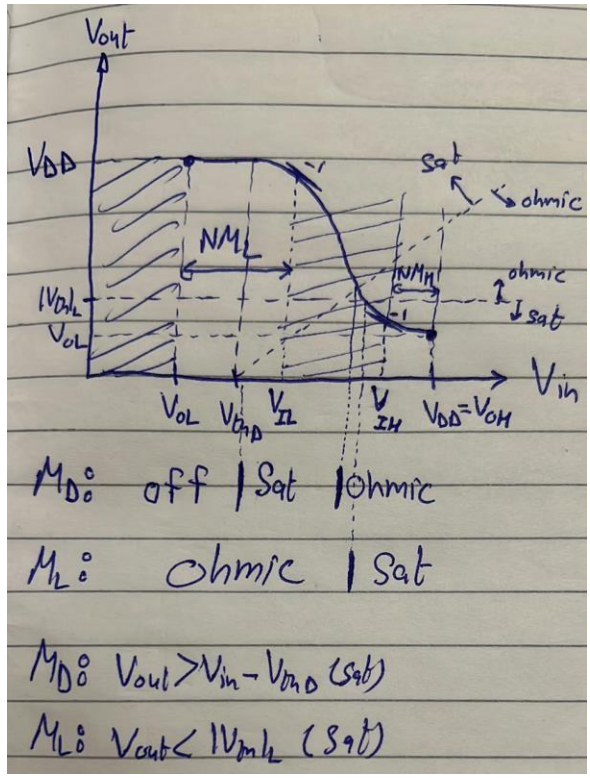
$V_{in}=V_{DD}$ ∴ M_D on. Sat or ohmic? we want it ohmic & assume ohmic. M_L? we want it sat & assume M_L in sat. V_{out} & I? KCL:

$$I_D|_{ohmic} = I_L|_{sat}.$$

$$k_D\left[(V_{DD}-V_{tD})(V_{OL})-\frac{V_{OL}^2}{2}\right] = \frac{k_L}{2}(V_{DD}-|V_{tL}|)^2 \Rightarrow \text{get } V_{OL}$$

verify your assumptions

2)



3) Same as previous question.